

ELEC3115 – Electromagnetic Engineering
Term 1, 2026 – Laboratory Information for Students

Laboratory Overview

This term, you will complete **THREE laboratory experiments**, scheduled **one every fortnight** during your regular 3-hour lab class.

The Experiments

1. **Experiment 1:** Permittivity Constant of Dielectrics
2. **Experiment 2:** AC Electromagnetic Induction and Inductance
3. **Experiment 3:** Eddy Currents, Hysteresis & Core Losses

Experiments **1 and 2** include **simulation-based pre-lab tasks**.

These **must be completed before attending your lab session**.

Attendance & Organisation

- All laboratory sessions are **compulsory**.
- Failure to complete any of the three labs may result in your **final course result being withheld**.
- All experiments are conducted in **Room EE205**.
- You will work in **pairs or groups of three**, with the same partners and in the same class time for all experiments.

There is **no structured catch-up session** during the term.

If you miss a lab, you must:

- Arrange an alternative session by contacting the head lab demonstrator.
- Provide appropriate documentation if absent due to medical reasons via UNSW special consideration.

A catch-up session may be arranged in **Week 11**, but only for approved medical absences.

Preparing for Your Lab

Good preparation makes the lab smoother and less stressful.

Before attending:

- Carefully read the **background theory**.
- Review the **experimental procedure**.
- Understand what measurements you will be making.

During the lab, you will assemble equipment in different configurations. Familiarising yourself with the manual beforehand will save valuable time.

Remember:

- Demonstrators will explain how to use equipment safely.
- They will not perform the experiment for you.
- Lab sessions are not lectures or tutorials.

You are expected to take responsibility for your own learning.

Demonstrators may not immediately correct non-dangerous mistakes. Learning from errors and refining your approach is part of the experimental process.

Assessment Overview

Laboratory work contributes **15% of your total course mark**.

Mark Distribution

- **Lab 1:** 40 marks (20 pre-lab + 20 in-lab)
 - **Lab 2:** 40 marks (20 pre-lab + 20 in-lab)
 - **Lab 3:** 20 marks (in-lab only)
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Pre-Lab Assessment (Labs 1 & 2 – 20 marks each)

You will complete an **ANSYS Maxwell simulation** at home and submit a **1–2 page report**.

Pre-Lab Rubric

- Correct results – 12 marks
- Oral explanation and interpretation – 8 marks

Due dates

- Pre-lab 1: 9 am, Monday (16 March), Week 5 via the specified submission channels in the Microsoft Teams page.
- Pre-lab 2: 9 am, Monday (30 March), Week 7 via the specified submission channels in the Microsoft Teams page.

In-Lab Assessment (20 marks per lab)

1. Completion of tasks & professional conduct – 4 marks
2. Well-presented lab book – 4 marks
3. Experimental skill – 6 marks
4. Understanding & interpretation (Viva Voce) – 6 marks

At the end of each session, your demonstrator will award marks for both pre-lab and in-lab components.

Recording Your Results

Developing good laboratory habits is essential.

Your lab book should:

- Clearly record what happened during the experiment.
- Include equipment settings and all measurements.
- Contain dated entries (start each new page with the date).

Important:

- Lab books must not consist of loose sheets.
- Printed plots must be securely glued in.
- If using a digital lab book, it must be submitted via the designated portal before the end of the lab session.

The teaching team reserves the right to inspect lab books before finalising marks.

Safety Information

These experiments involve standard electrical hazards.

You must:

- Wear closed shoes.
- Keep food and drinks away from lab benches.
- Follow all safety instructions given by demonstrators.

Be aware that:

- Some components become hot during operation.
- Labs can be crowded — avoid creating trip hazards.

Students who persist in unsafe behaviour may be removed from the lab session (which may result in forfeited marks).

Contact Information

Head Lab Demonstrator: Ravneel Prasad ravneel.prasad@unsw.edu.au

Lecturer in Charge: Rukmi Dutta rukmi.dutta@unsw.edu.au

Lab Technical Officer: R. M. Gamini Liyadipitiya (Gamini) g.liyadipitiya@unsw.edu.au

We look forward to working with you in the laboratory.
Careful preparation, active participation, and professional conduct will ensure you gain the maximum benefit from these sessions.